

WOLF POINT CLAYTON, MT, USA

WMO: 727686

Lat: 48.094N

Lon: 105.574W

Elev: 1986

StdP: 13.67

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94017

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.4	-15.0	-26.5	1.3	-20.3	-20.5	1.9	-13.9	28.3	24.3	25.6	23.5	5.2	250	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.8	95.5	66.0	91.9	65.6	88.4	64.7	70.7	86.7	68.5	84.5	66.7	83.2	9.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	101.5	78.7	63.3	93.6	75.1	61.2	86.8	73.0	35.9	86.5	33.9	84.0	32.3	82.9	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.2	23.0	19.6	-29.7	101.2	5.9	3.2	-33.9	103.5	-37.3	105.4	-40.6	107.2	-44.9	109.5		
			-28.8	74.6	4.5	2.8	-32.1	76.6	-34.7	78.2	-37.3	79.8	-40.6	81.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	14.2	15.6	30.3	43.7	53.9	63.3	71.8	69.3	58.2	44.2	29.2	16.3	
	DBStd	22.81	14.21	14.11	13.58	8.95	8.35	6.66	5.99	6.51	8.31	8.95	12.15	13.38	
	HDD50	4876	1110	963	613	224	55	2	0	0	21	218	626	1044	
	HDD65	8641	1575	1383	1074	640	354	108	12	32	234	645	1075	1509	
	CDD50	2194	0	0	4	34	176	400	675	597	268	39	1	0	
	CDD65	483	0	0	0	0	10	56	221	164	31	1	0	0	
	CDH74	7909	0	0	2	53	322	1007	3166	2547	755	57	0	0	
	CDH80	3564	0	0	0	9	79	348	1561	1241	314	12	0	0	
Wind	WSAvg	8.1	7.6	7.4	8.8	10.0	9.5	8.5	7.8	7.6	7.5	8.0	7.4	7.3	
Precipitation	PrecAvg	14.0	0.4	0.2	0.5	1.0	2.1	3.2	2.1	1.7	1.0	0.8	0.4	0.3	
	PrecMax	19.6	1.4	0.7	1.5	2.5	5.6	6.5	6.1	7.1	3.4	3.3	3.1	0.8	
	PrecMin	6.4	0.0	0.0	0.0	0.3	0.4	1.2	0.4	0.0	0.1	0.0	0.0	0.0	
	PrecStd	3.0	0.3	0.2	0.4	0.6	1.3	1.4	1.3	1.6	0.9	0.7	0.6	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	52.1	71.4	80.6	87.3	93.4	100.3	100.1	94.2	81.3	65.5	48.7	
		MCWB	39.5	41.6	51.2	55.3	61.9	65.7	67.4	67.1	64.8	57.4	49.5	39.2	
	2%	DB	40.7	44.0	62.7	72.8	81.8	88.2	96.6	95.1	88.3	72.4	57.4	42.6	
		MCWB	35.4	36.9	47.0	52.8	59.6	66.0	67.4	65.4	62.0	54.2	44.9	36.1	
	5%	DB	37.1	38.7	55.4	67.5	77.5	83.8	92.9	91.2	82.8	66.1	51.8	37.7	
		MCWB	33.5	34.2	43.1	49.6	57.6	63.1	66.7	64.7	60.4	51.0	42.1	33.4	
	10%	DB	34.0	34.8	49.6	62.9	72.6	80.0	89.3	87.6	77.0	61.1	45.9	34.3	
		MCWB	31.0	31.8	40.6	47.5	55.2	62.0	66.1	63.8	58.1	48.4	38.6	31.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	41.9	51.5	57.0	64.8	71.9	75.2	71.4	67.8	59.7	50.4	39.7
			MCDB	46.4	50.9	69.6	76.1	80.2	85.9	91.0	91.2	83.5	77.8	63.2	46.8
2%		WB	36.0	37.3	48.0	53.3	61.4	68.3	71.9	68.6	64.3	55.3	46.1	36.3	
		MCDB	40.2	43.2	62.1	71.0	77.4	82.8	87.5	86.2	82.6	68.1	57.2	41.8	
5%		WB	33.6	34.5	44.1	50.6	59.2	65.8	69.6	67.0	61.7	52.0	42.2	33.8	
		MCDB	37.1	38.6	54.6	65.2	74.4	79.6	85.1	85.2	79.7	64.5	50.5	37.6	
10%		WB	31.1	31.8	41.0	48.3	56.7	63.4	67.8	65.3	59.2	49.4	39.1	31.4	
		MCDB	34.0	35.0	48.8	61.1	70.0	76.6	83.6	83.5	75.6	60.2	45.5	34.6	
Mean Daily Temperature Range	5% DB	MDBR	20.9	21.8	23.4	27.0	27.0	26.1	29.8	30.6	30.1	26.2	23.8	20.8	
		MCDBR	22.8	26.5	34.2	37.9	35.1	34.1	35.7	37.6	40.3	37.2	33.1	25.3	
		MCWBR	18.5	20.3	20.6	20.0	16.9	15.1	13.1	14.0	17.9	20.7	21.3	19.7	
	5% WB	MCDBR	22.6	25.1	32.5	34.9	31.6	29.3	30.2	31.3	36.2	34.7	31.6	24.2	
		MCWBR	18.5	19.4	19.9	18.9	15.8	14.4	12.9	12.8	16.5	19.7	20.8	19.2	
Clear-Sky Solar Irradiance	taub	0.254	0.263	0.278	0.321	0.342	0.355	0.363	0.372	0.325	0.290	0.262	0.245		
	taud	2.486	2.511	2.547	2.425	2.416	2.428	2.420	2.365	2.467	2.560	2.542	2.518		
	Ebn at Noon	270	292	303	296	292	288	283	275	280	277	265	261		
	Edh at Noon	19	23	27	34	35	35	35	36	29	22	18	16		
All-Sky Solar Radiation	RadAvg	470	791	1185	1566	1865	2056	2203	1806	1323	840	508	358		
	RadStd	40	56	81	78	154	123	96	120	115	79	56	20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page